Reconstructing Digital Archives Through Full Stack Feminism: A Case Study of Women in Science

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Introduction

The Full Stack Feminism *in Digital Humanities* (FSF) project defines digital humanities not merely as the application of digital tools to humanities research but as a practice that integrates social, cultural, and technological critique across all levels of digital infrastructure. According to its project statement, FSF aims to "develop an intersectional feminist framework for digital arts and humanities practice and research" (Webb et al. 2023). This framework is built on the idea that feminist intervention should occur throughout the "Full Stack" of digital systems, from the code and data infrastructure to the frontend and interfaces that shape users' experience.

To illustrate how the principles of Full Stack Feminism operate in practice, this essay conducts a comparative analysis of FSF's framework through a close reading of The History of Women in Science: A Resource of the American Philosophical Society (American Philosophical Society 2023). The Women in Science project is a digital archive that documents the contributions of women scientists who were historically excluded or marginalized within mainstream scientific institutions. It provides digitized manuscripts, correspondence, and visual materials that make women's efforts visible across the history of science.

Analyzing the digital archive through Full Stack Feminism reveals how feminism is applied in digital humanities research and challenges traditional patriarchy. Women in Science succeeds in increasing accessibility so that people have the opportunity to learn more about the disappeared women in science. A full-stack feminist critique also draws attention to metadata structures and user interfaces. By applying FSF's intersectional and ethical framework, this study examines how

feminist ideas, histories, and data about women in science are created, represented, and circulated within digital infrastructures.

Principles of Full Stack Feminism

Full Stack Feminism emphasizes that digital projects are not neutral technical objects but contain social and cultural meaning. The feminist digital humanities should include coding practices, metadata design, and interface presentation to uncover how systemic biases are encoded (Webb et al. 2023). This perspective challenges the assumption that digital tools are a neutral means to spreading knowledge, instead highlighting the need for critical attention to who is included or excluded in digital representations.

A core principle of FSF is intersectionality, which focuses on overlapping forms of marginalization. Digital archives and platforms must account for multiple identities, such as race, class, gender, and disability, when arranging content or designing user interactions (Webb et al. 2023). This requires transparent and collaborative design processes, ensuring that decisions about data structuring, categorization, and presentation are guided by ethical and inclusive considerations.

FSF also pays attention to community-led design, actively involves the communities it represents when creating content and platforms. Diverse communities have access to demonstrate their experiences and knowledge through storytelling, and digital infrastructures make the speech visible, and materialization reflect feminist values (Trinh 1989, 121). This principle is particularly relevant for historical archives of marginalized groups, as it encourages collaboration, inclusive storytelling, and attention to how material and digital forms intersect to shape knowledge.

In general, FSF emphasizes the importance of feminist infrastructure as both a technical and cultural project. By analyzing the three stacks, (1) data and archives, (2) infrastructure, tools, and code, (3) access, experience, and integration, scholars can assess how digital systems either reproduce or challenge patriarchal epistemologies. This framework provides the theoretical foundation for evaluating digital archives such as The History of Women in Science, allowing for critical thinking in both content and structural design.

Case Study Analysis: Women in Science at the American Philosophical Society

The Women in Science project at the American Philosophical Society (APS) presents the visual archive of women scientists by mining correspondence and network data. As described on its "About" page, the archive identifies biographical and network bonds by surveying letters within APS collections and visualizing the connections between individuals (Sutherland, Nelson, and Miller 2024).

One critical feature of the project is its public acknowledgment of the shortcomings of the archives. The creators note that the dataset includes only those scientists represented in the APS's existing holdings; some influential figures, such as Maria Mitchell, are absent "only because she was not referred directly within the surveyed collections" (Sutherland, Nelson, and Miller 2024). This transparency regarding the missing information highlights the consideration of how data collection can shape historical narratives. It reveals a feminist consciousness of absence, recognizing that what is missing from the archive is as important as what is included.

The technical structure of Women in Science demonstrates how data infrastructures influence meaning. The archive evolved through several development phases: data was gathered by Serenity Sutherland during her digital humanities fellowship, followed by the restructuring of formats from CSV to JSON, and then visualized by David Ragnar Nelson and Bayard L. Miller using JavaScript-related tools (Sutherland, Nelson, and Miller 2024). How data are modeled, formatted, and visualized, such design choices embody Full Stack Feminism, as Webb argue that "It involves reframing and reimaging the history of computing…through an intersectional, queer, anti-racist, lens challenges the heteronormative, masculinized, and neo-liberal structures which facilitate and perpetuate inequity whether manifested in binary code, or social, cultural and political ones" (2023). These practices reveal which relationships are made visible and which remain obscured, reminding users that visualization is an act rather than a neutral narrative.

The project redefines the recovery of women's histories as an ongoing, reflective process rather than merely adding missing names. It recognizes that archival and data limitations influence what can be shown, making recovery an imperfect and partial process with digital evidence. By

emphasizing relationships over isolated achievements, Women in Science challenges the idea of exceptional women in the patriarchal story and instead underscores the collective role of women in science. However, the project's dependence on existing institutional archives means that intersectional aspects like race, class, and geography are still underrepresented. This highlights the ongoing need for feminist infrastructural theory and critique, aligning with Webb's call to "disrupt or challenge the normative and sometimes archaic modes of object and subject descriptions in catalogs" (2023).

Interface Design and User Experience

In the context of Full Stack Feminism (FSF), user experience and interface design represent the digital feminist becoming visible material. As Webb explains, FSF identifies "access, experience, and integration" as one of its three key stacks, emphasizing that the feminist critique should extend to the ways users encounter and interact with digital projects (Webb 2023). This perspective lets people consider how to resist hierarchies embedded in digital infrastructures.

When analyzing The History of Women in Science through this framework, the archive's design prioritizes accessibility and clarity rather than authority or exclusivity. The website's organization and metadata structure allow users to explore women's contributions through diverse paths, for example, topic, geography, or correspondence networks, thus embodying a feminist approach to researching a project. Its interface avoids classification systems that traditionally privilege certain disciplines or figures. This reflects FSF's broader goal of "centering ethics of care and inclusivity across all layers of the stack" (Webb 2023).

FSF emphasizes that the ways users engage with archives are shaped by the underlying design and infrastructure (Webb 2023). In the Women in Science archive, this principle is reflected through the relational visualization of women scientists' networks. By constructing visualization around connections and correspondence rather than hierarchical or linear lists, the archive encourages users to engage actively with historical relationships and to consider the social contexts that shaped women's scientific work. This approach aligns with FSF's insistence that user experience is not neutral but connected with ethical and feminist considerations. The

meaning of navigating digital archives as a critical act is far more than receiving information without actively thinking and interacting with it (Webb 2023).

Conclusion

Through the learning of Full Stack Feminism and the History of Women in Science, these demonstrate how feminist principles can be included in data, infrastructure, design, and user experience. By thinking about access and interaction, the project extends the feminist intervention from the invisible code layer to the visible interface, where users receive information. As Webb said, FSF aims to "embed feminist value through every layer of digital stack" (2023), and Women in Science offers a specific realization of this vision. Digital engagement is shaped as an inclusive act that diverse communities can participate in, ensuring that the women once marginalized by traditional scientific discourse are not only remembered but also presented in a different way than tradition within digital space.

Reference

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